

Statistical Learning Notes

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1 Intro to Multivariate Analysis

1.1 Multivariate

Multivariate data sets are several observations are made on each individual or object in one or more samples. We analyse it because:

- Investigations collect data simultaneously on many variables that are relevant
- Univariate analysis of such data does not convey the complete story
- Multivariate analysis attempts to provide methods that will concurrently analyse such data

1.2 Common Aims

1.2.1 Interdependence

UNSUPERVISED (no dependent or independent variables)

Data reduction or structural simplification: Attempt to represent the multivariate data in a simple possible way without losing valuable information and seek easier interpretation of the phenomenon → *Principal Component Analysis*

Sorting and grouping: Construct from the multivariate data rules that can be used to classify an object into one of the well defined groups → *Cluster Analysis*

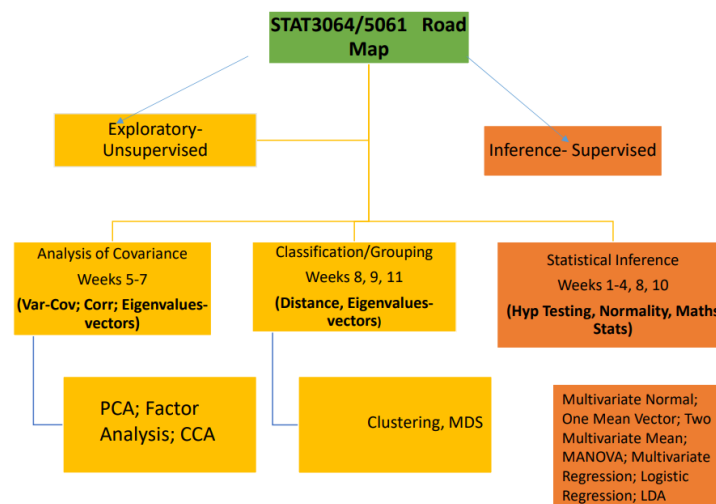
1.2.2 Dependence

SUPERVISED (dependent and independent variables)

Dependence among variables: Investigate relationships (if present) between relevant variables of the multivariate data → *Multivariate Regression Analysis*

Prediction: attempt to construct a prediction method for one or more variables from a related set of observed variables

Hypothesis Testing: construct methods of testing a postulated hypothesis → *mean vector; Two Multivariate Mean; MANOVA*



2 Multivariate Data

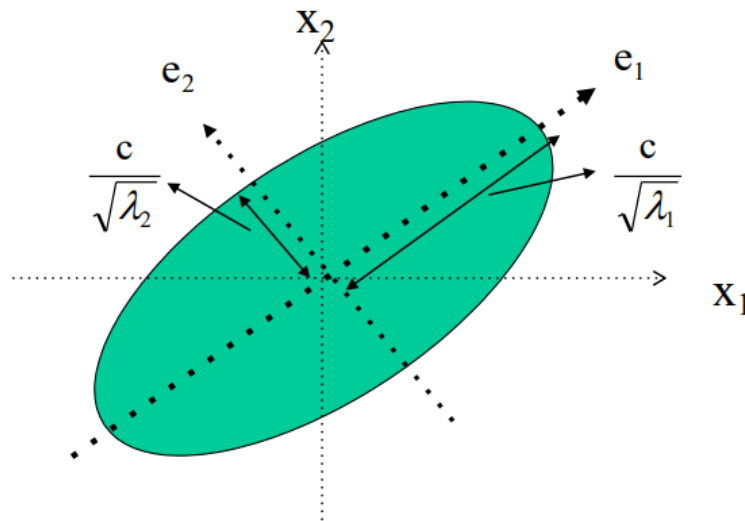
2.1 Spectral Decomposition

$$A = \lambda_1 e_1 e_1' + \lambda_2 e_2 e_2' + \dots + \lambda_k e_k e_k'$$

2.2 Positive Definite Matrices and Eigenvalues - A symmetric matrix A of order k is non-negative definite matrix if all of its eigenvalues are greater than or equal to zero - A symmetric matrix A of order k is positive definite matrix if all of its eigenvalues are greater than zero

2.3 Statistical Distance

$$(d(P, O))^2 = c^2 = X'AX = \lambda_1 y_1^2 + \lambda_2 y_2^2$$



2.4 Correlation and Covariance

$$V^{1/2} \rho V^{1/2} = \sigma$$

$$\rho = (V^{1/2})^{-1} \sigma (V^{1/2})^{-1}$$

2.4.1 Generalised Sample Variance

An unbiased estimator of Σ is:

$$E\left(\frac{n}{n-1} S_n\right) = \Sigma$$

R does this already with:
var(data)

2.5 Statistical Independence

- X_i and X_k are statistically independent if for all pairs of values (x_i, x_k) we have:

$$P[X_i \leq x_i, X_k \leq x_k] = P[X_i \leq x_i]P[X_k \leq x_k]$$

- When X_i and X_k are continuous random variables, they are statistically independent if for all pairs of values (x_i, x_k) , the joint density is equal to the product of the marginal densities:

$$f_{ik}(x_i, x_k) = f_i(x_i)f_k(x_k)$$

- As a result statistical independence *implies* that:

$$\text{Cov}(X_i, X_k) = \sigma_{ik} = 0$$

- But the converse is not true in general (see 12.2)

2.6 Linear Combination

- Let $\mathbf{c}' = [a, b]$, $\mathbf{X}' = [X_1, X_2]$, $\boldsymbol{\mu}' = [\mu_1, \mu_2]$
- and $\text{cov}(\mathbf{X}) = \Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{bmatrix}$

Now

$$E(aX_1 + bX_2) = E(\mathbf{c}'\mathbf{X}) = a\mu_1 + b\mu_2 = \mathbf{c}'\boldsymbol{\mu}$$

$$\text{Var}(aX_1 + bX_2) = \text{Var}(\mathbf{c}'\mathbf{X}) = a^2\sigma_{11} + b^2\sigma_{22} + 2ab\sigma_{12} = \mathbf{c}'\Sigma\mathbf{c}$$

$$\text{Cov}(aX_1, bX_2) = ab\sigma_{12} \rightarrow \text{Cov}(CX) = \mathbf{C}\Sigma\mathbf{C}'$$

3 Univariate and Multivariate Normal Distribution

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)(\sigma^2)^{-1}(x-\mu)/2}$$

3.1 Maximum Likelihood Estimate (MLE)

Likelihood is joint density or frequency function

- Find the value of θ that maximises the likelihood (make the data most probable or likely)
- If X's are iid, then:

$$lik(\theta) = \prod_{i=1}^n f(x_i|\theta)$$

- Easier to work with log likelihood:

$$l(\theta) = \sum_{i=1}^n \log[f(x_i|\theta)]$$

- Maximise by differentiating $lik(\theta)$ or $l(\theta)$ and setting to zero

Joint likelihood:

$$f(x_1, x_2, \dots, x_n | \mu, \sigma) = \prod_{i=1}^n \frac{1}{\sigma \sqrt{2\pi}} \exp \left(-\frac{1}{2} \left(\frac{x_i - \mu}{\sigma} \right)^2 \right)$$

Log likelihood:

$$l(\mu, \sigma) = -n \log \sigma - \frac{n}{2} \log 2\pi - \frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2$$

First derivative of μ and σ : $\frac{\partial l}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \mu)$, $\frac{\partial l}{\partial \sigma} = -\frac{n}{\sigma} + \sigma^{-3} \sum_{i=1}^n (X_i - \mu)^2$

MLE:

$$\hat{\sigma} = \sqrt{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2}$$

$$\hat{\mu} = \bar{X}$$

3.2 Good estimators

Unbiased: - $E(\hat{\theta}) = \theta$ - Estimator is ‘targetted’ to the desired value

Consistent: - $Var(\hat{\theta}) \rightarrow 0$ as $n \rightarrow \infty$ - The estimator becomes more ‘reliable’ (less variable) as the sample size increases

3.2.1 Bivariate Normal

For $X \sim N(\mu_X, \sigma_X^2)$, the marginal distribution of X is:

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma_X} \exp \left(-\frac{1}{2} \frac{(x - \mu_X)^2}{\sigma_X^2} \right)$$

For X, Y bivariate normal:

$$f(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y} \exp \left(-\frac{1}{2(1-\rho^2)} \left[\frac{(x - \mu_X)^2}{\sigma_X^2} + \frac{(y - \mu_Y)^2}{\sigma_Y^2} - \frac{2\rho(x - \mu_X)(y - \mu_Y)}{\sigma_X\sigma_Y} \right] \right)$$

Where $-1 < \rho < 1$ (the last term represents the correlation coefficient, ρ).

3.2.2 Conditional Distribution

Here we use the property that the conditional distribution is also normal. Recall that:

$$\mathbf{X}_1 | \mathbf{X}_2 = x_2 \sim N(\mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2), \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}).$$

Hence:

$$\mathbf{X}_2 | \mathbf{X}_1 = x_1 \sim N(\mu_2 + \Sigma_{21}\Sigma_{11}^{-1}(x_1 - \mu_1), \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}).$$

Conditional Distribution

```
conMSon=means[2,1]+sigma[2,1]*(sigma[1,1])^(-1)*(185-means[1,1])
convar=sigma[2,2] - sigma[2,1]*(sigma[1,1])^(-1)*sigma[1,2]
## P(X2 < 185 | X1=185)
pnorm(185,conMSon,sqrt(convar))
```


3.3 Multivariate Normal Distribution

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (\mathbf{x} - \mu)' \Sigma^{-1} (\mathbf{x} - \mu) \right)$$

, where $-\infty < x_i < \infty$, $1 \leq i \leq p$.

This p -dimensional normal density is denoted by $N_p(\mu, \Sigma)$.

3.3.1 The Hyper ellipsoid

$$(\mathbf{X} - \mu)' \Sigma^{-1} (\mathbf{X} - \mu) = c^2$$

This hyperellipsoid is **centered** at μ and has **axes** $\pm c\sqrt{\lambda_i} \mathbf{e}_i$, where $(\lambda_i, \mathbf{e}_i)$ is an eigenvalue and eigenvector pair of Σ . That is,

$$\Sigma \mathbf{e}_i = \lambda_i \mathbf{e}_i, \quad \text{for } i = 1, 2, \dots, p.$$

3.4 Properties of Multivariate Normal

- Linear combination of components of $\mathbf{X}$ are normal
- All subsets of components of $\mathbf{X}$ have a multivariate normal distribution
- Zero covariance implies corresponding components are independently distributed
- The conditional distribution of components are multivariate normal

3.4.1 Linear Combination of Components of a Normal Random Vector

1. If $\mathbf{X}$ is distributed as $N_p(\mu, \Sigma)$, then any linear combination of the p variables, $\mathbf{a}'\mathbf{X}$ is distributed as $N(\mathbf{a}'\mu, \mathbf{a}'\Sigma\mathbf{a})$.
2. Given a random vector $\mathbf{X}$, a column vector μ , and a symmetric matrix Σ of appropriate dimension, if $\mathbf{a}'\mathbf{X}$ is distributed as $N(\mathbf{a}'\mu, \mathbf{a}'\Sigma\mathbf{a})$ for every $\mathbf{a}$, then $\mathbf{X}$ follows $N_p(\mu, \Sigma)$.
3. If random vector $\mathbf{X}$ follows $N_p(\mu, \Sigma)$, and A is a $q \times p$ matrix, then $A\mathbf{X}$ follows $N_q(A\mu, A\Sigma A')$. Furthermore, if $\mathbf{d}$ is a $p \times 1$ vector of constants, then $\mathbf{X} + \mathbf{d}$ follows $N_p(\mu + \mathbf{d}, \Sigma)$.
4. If random vector $\mathbf{X}$ follows $N_p(\mu, \Sigma)$, then **every subset** of $\mathbf{X}$ is also normally distributed.
5. If $\mathbf{X}^{(1)'} = [X_1, X_2, \dots, X_q]$ and $\mathbf{X}^{(2)'} = [X_{q+1}, X_{q+2}, \dots, X_p]$ are **independent**, then

$$\text{Cov}(\mathbf{X}^{(1)}, \mathbf{X}^{(2)}) = \mathbf{0}_{q \times (p-q)}.$$

6. $\mathbf{X}^{(1)}$ and $\mathbf{X}^{(2)}$ are independent **if and only if** $\Sigma_{12} = 0$.
7. If $\mathbf{X}^{(1)}$ and $\mathbf{X}^{(2)}$ are not independent and $|\Sigma_{22}| > 0$, then the conditional distribution of $\mathbf{X}^{(1)}$ given $\mathbf{X}^{(2)} = \mathbf{x}^{(2)}$ is normal with mean:

$$\mu^{(1)} + \Sigma_{12}(\Sigma_{22})^{-1}(\mathbf{x}^{(2)} - \mu^{(2)})$$

and covariance:

$$\Sigma_{11} - \Sigma_{12}(\Sigma_{22})^{-1}\Sigma_{21}.$$

8. If $\mathbf{X} \sim N_p(\mu, \Sigma)$ with $|\Sigma| > 0$, then:

- (a) $(\mathbf{X} - \mu)' \Sigma^{-1} (\mathbf{X} - \mu)$ is distributed as χ_p^2
- (b) The probability that $(\mathbf{X} - \mu)' \Sigma^{-1} (\mathbf{X} - \mu) \leq \chi_p^2(\alpha)$ is $1 - \alpha$, where $\chi_p^2(\alpha)$ is the upper 100α -th percentile.

3.4.1 Testing Normality and Confidence Interval

Recollect that:

$$\text{Prob} \{ \mathbf{x} : (\mathbf{x} - \mu)' \Sigma^{-1} (\mathbf{x} - \mu) \leq \chi_p^2(\alpha) \} = 1 - \alpha$$

where $\chi_p^2(\alpha)$ is the upper 100α -th percentile.

Now, for any subset of bivariate outcomes, say (x_{ji}, x_{jk}) , $j = 1, \dots, n$, we can look at the scatter and expect 50% of the sample observations to lie within the ellipse given by:

$$\left(\begin{pmatrix} x_{ji} \\ x_{jk} \end{pmatrix} - \begin{pmatrix} \bar{x}_i \\ \bar{x}_k \end{pmatrix} \right)' \begin{pmatrix} s_{ii} & s_{ik} \\ s_{ki} & s_{kk} \end{pmatrix}^{-1} \begin{pmatrix} x_{ji} \\ x_{jk} \end{pmatrix} - \begin{pmatrix} \bar{x}_i \\ \bar{x}_k \end{pmatrix} \leq \chi_2^2(0.5)$$

3.4.2 QQ Plot / Chi-square Plot

Consider the squared distances:

$$d_j^2 = (\mathbf{x}_j - \bar{\mathbf{x}})' S^{-1} (\mathbf{x}_j - \bar{\mathbf{x}}), \quad 1 \leq j \leq n$$

where $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ are the sample observations.

- Degrees of freedom = p - n-p need to be greater than 25
- We can construct a chi-square plot with these squared distances and inspect for a straight line plot through the origin

1. Arrange squared distances such that:

$$d_{(1)}^2 \leq d_{(2)}^2 \leq \dots \leq d_{(n)}^2.$$

2. Plot the pairs $\left(q_{c,p} \left(\frac{j-1/2}{n} \right), d_{(j)}^2 \right)$, for $j = 1, 2, \dots, N$, where:

$$q_{c,p} \left(\frac{j-1/2}{n} \right) = \chi_p^2 \left(\frac{n-j+1/2}{n} \right).$$

- We expect the plot to be a straight line passing through the origin with slope 1.
- Otherwise, we suspect a lack of normality.

```
# Ellipsoid
eigS <- eigen(Sigma)
vals <- eigS$val
vec <- eigS$vec
c <- sqrt(qchisq(0.5,df=2))
axes <- c * sqrt(vals)

axes[1]*vec[1,1]
axes[1]*vec[2,1]+mu[2]

axes[2]*vec[1,2]
axes[2]*vec[2,2]+mu[2]

# Libraries
library(MVN)
library(MASS)
```

```

# QQ plot
mvn(data= setosa, multivariatePlot = "qq")

mvn(data= setosa, mvnTest="royston", multivariateOutlierMethod="quan")
mvn(data=versicolor,mvnTest="hz",multivariateOutlierMethod="adj")

qqplot(qchisq(ppoints(500),df=p), d, main="", xlab="Theoretical Quantiles", ylab="Sample Quantiles")
qqline(d,distribution=function(x){qchisq(x,df=p)})

# MuN Test
mvn(data= setosa,mvnTest="mardia")
mvn(data= setosa,mvnTest="hz")
mvn(data= setosa,mvnTest="royston")
mvn(data= setosa,mvnTest="dh")
mvn(data= setosa,mvnTest="energy")

# Distance
d <- diag(t(t(X)-Xbar)%*%Sinv%*%(t(X)-Xbar))

```

4 Inference For Multivariate Data

4.1 Hypothesis Testing

6 Steps to Hypothesis Testing

1. State Hypothesis
2. Calculate the Test Statistic
3. Determine Sampling Distribution of the Test Statistic (2)
4. Find the p-value (3)
5. Make a decision based on 4 at α significance
6. State your conclusion in the context of your specific setting

4.1.1 Univariate T-Test

Student's t-test follows t-distribution

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \sim T_{n-1}$$

With *Degrees of Freedom* = $n - 1$

4.1.2 p-variate Test

$$T^2 = n(\bar{X} - \mu_0)'S^{-1}(\bar{X} - \mu_0)$$

4.1.3 Hotelling's Test

$$n(\bar{X} - \mu)'S^{-1}(\bar{X} - \mu) > \frac{(n-1)p}{n-p}F_{p,n-p}(\alpha)$$

```
library(DescTools)
HotellingsT2Test(data, mu=c(0,0,0))

d=eff[1:11,3:4]-eff[12:22,3:4] # Differences between Comm-State for each bod and ss
HotellingsT2Test(d)

t2 = n * t(mu) %*% solve(S) %*% mu
t2h = (t2 *(n-p))/(p *(n-1))
fcrit=(3*18/16)*qf(0.95,3,16)
pval=pf(34,3,16, lower.tail = F)
```

4.2 Confidence Region

For a given random sample $X_1, X_2, \dots, X_n$ from a normal distribution with unknown Σ and μ , the $100(1-\alpha)\%$ confidence region for the mean is the ellipsoid defined by the set of all μ such that:

$$n(\bar{X} - \mu)'S^{-1}(\bar{X} - \mu) \leq \frac{(n-1)p}{n-p}F_{p,n-p}(\alpha)$$

4.2.1 Axes of confidence ellipsoid and their relative lengths

λ_i and e_i are the eigenvalue and eigenvector for S for $i=1, 2, \dots, p$

Let $c^2 = \frac{(n-1)p}{n-p} F_{p, n-p}(\alpha)$

Centre of Ellipsoid is: $\bar{X}$

Direction of Axis: e_i

Length of Axis: $\sqrt{\lambda_i}c/\sqrt{n}$

4.3 Simultaneous Confidence Intervals

If we want to know the simultaneous confidence interval for μ_1 :

$$\bar{X}_1 - \sqrt{\frac{p(n-1)}{n-p} F_{p, n-p}(\alpha)} \sqrt{\frac{s_{11}}{n}} \leq \mu_1 \leq \bar{X}_1 + \sqrt{\frac{p(n-1)}{n-p} F_{p, n-p}(\alpha)} \sqrt{\frac{s_{11}}{n}}$$

```
sim.int <-function(xbar, p, n, alpha, vars){
  fval=((n-1)*p/(n-p))*qf(alpha,p,n-p, lower.tail = FALSE)
  me=sqrt(fval*vars/n)
  lower=xbar-me
  upper=xbar+me
  return(c(lower,upper))
}
```

4.4 Simultaneous confidence Statements

The linear combination $Z = a'X$ follows a $N(a'\mu, a'\Sigma a)$ distribution. Confidence interval for $a'\mu$ is:

$$a'\bar{X} - \sqrt{\frac{p(n-1)}{n(n-p)} F_{p, n-p}(\alpha)} a'Sa \leq a'\mu_1 \leq a'\bar{X} + \sqrt{\frac{p(n-1)}{n(n-p)} F_{p, n-p}(\alpha)} a'Sa$$

4.5 Bonferroni Corrections

The simultaneous confidence intervals for components of μ_k are:

$$\left(x_k - t_{n-1} \left(\frac{\alpha}{2p} \right) \sqrt{\frac{s_{kk}}{n}}, x_k + t_{n-1} \left(\frac{\alpha}{2p} \right) \sqrt{\frac{s_{kk}}{n}} \right), k = 1, 2, \dots, p$$

```
bon.int <-function(xbar, p, n, alpha, vars){
  tval=qt((alpha/(2*p)), n-1, lower.tail = F)
  me=tval*sqrt(vars/n)
  lower=xbar-me
  upper=xbar+me
  return(c(lower,upper))
}
```

4.6 Homoscedasticity

Equality of covariance matrix $\Sigma_1 = \Sigma_2 = \Sigma$

$$S_{pooled} = \frac{(n_1 - 1)S_1 + (n_2 - 1)S_2}{n_1 + n_2 - 2}$$

The difference of sample means follows:

$$\bar{X}_1 - \bar{X}_2 \sim N_p \left(\mu_1 - \mu_2, \left(\frac{1}{n_1} + \frac{1}{n_2} \right) \Sigma_{pooled} \right)$$

The test statistic T^2 is given by:

$$T^2 = (\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2))' \left[\left(\frac{1}{n_1} + \frac{1}{n_2} \right) S_{pooled} \right]^{-1} (\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)) \sim \frac{(n_1 + n_2 - 2)p}{(n_1 + n_2 - p - 1)} F_{p, n_1 + n_2 - p - 1}$$

Confidence Interval For any given $a = (a_1, \dots, a_p)'$, the confidence interval for $a'(\mu_1 - \mu_2)$ at the $1 - \alpha$ level is:

$$a'(\bar{X}_1 - \bar{X}_2) \pm c \left[a' \left(\frac{1}{n_1} + \frac{1}{n_2} \right) S_{pooled} a \right]^{1/2}$$

Where $c^2 = \frac{(n_1 + n_2 - 2)p}{(n_1 + n_2 - p - 1)} F_{p, n_1 + n_2 - p - 1}(\alpha)$

Simultaneous CI:

$$(\bar{X}_{1i} - \bar{X}_{2i}) \pm c \left[\left(\frac{1}{n_1} + \frac{1}{n_2} \right) S_{ii, pooled} \right]^{1/2}, \quad i = 1, 2, \dots, p$$

Bonferroni CI:

$$(\bar{X}_{M,i} - \bar{X}_{F,i}) \pm t_{n_1 + n_2 - 2, \frac{\alpha}{2p}} \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2} \right) S_{ii, pooled}}$$

```
Spooled=((nf-1)/(nf+nm-2))*SF+((nm-1)/(nf+nm-2))*SM
```

4.6 Heteroscedasticity

Inequality of covariance matrix $\Sigma_1 \neq \Sigma_2$ Follows:

$$\bar{X}_1 - \bar{X}_2 \sim N_p \left(\mu_1 - \mu_2, \frac{1}{n_1} \Sigma_1 + \frac{1}{n_2} \Sigma_2 \right)$$

Test statistic T^2 For large n_1 and n_2 :

$$T^2 = (\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2))' \left(\frac{S_1}{n_1} + \frac{S_2}{n_2} \right)^{-1} (\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)) \sim \chi_p^2$$

Simultaneous CI:

$$(\bar{X}_{1i} - \bar{X}_{2i}) \pm (\chi_p^2(\alpha))^{1/2} [n_1^{-1} S_{ii,1} + n_2^{-1} S_{ii,2}]^{1/2}, \quad i = 1, 2, \dots, p$$

4.7 Difference in Population Hypothesis Testing

δ is the population vector mean of
Null hypothesis zero differences
 $H_0 : \delta = (\delta_1, \dots, \delta_p)' = (0, \dots, 0)'$.

Reject $H_0 : \delta = 0$, at α significance level, if

$T^2 = n\bar{\mathbf{D}}'(\mathbf{S}_D)^{-1}\bar{\mathbf{D}} > [(n-1)p/(n-p)]F_{p,n-p}(\alpha),$
 $F_{p,n-p}(\alpha)$ is the upper (100α) th percentile of $F_{p,n-p}$.

Write a R-program for calculation in Example 7

Confidence Region and Simultaneous confidence intervals

— Confidence region:

$$(\bar{\mathbf{D}} - \delta)'(\mathbf{S}_D)^{-1}(\bar{\mathbf{D}} - \delta) \leq [(n-1)p/\{n(n-p)\}]F_{p,n-p}(\alpha).$$

— simultaneous confidence interval:

$$\delta_i : \bar{d}_i \pm \sqrt{[(n-1)p/(n-p)]F_{p,n-p}(\alpha)} \sqrt{s_{d_i}^2/n}.$$

— Bonferroni simultaneous confidence interval:

$$\delta_i : \bar{d}_i \pm t_{n-1}(\alpha/(2p)) \sqrt{s_{d_i}^2/n}.$$

5 Principal Component Analysis

5.1 Purpose

Explaining the variance-covariance structure of a set of variables through a few linear combinations of these variables.

General Objectives are:

- Data reduction
- Interpretation
- Uncorrelated
- First few contain most of the variation

5.2 PCA with Sigma

Let Σ be the covariance matrix associated with the random vector $\mathbf{X}' = [X_1, X_2, \dots, X_p]$. Let Σ have the eigenvalue-eigenvector pairs $(\lambda_1, \mathbf{e}_1), (\lambda_2, \mathbf{e}_2), \dots, (\lambda_p, \mathbf{e}_p)$ where $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p \geq 0$.

$$Y_i = \mathbf{e}_i' \mathbf{X} = e_{i1}X_1 + e_{i2}X_2 + \dots + e_{ip}X_p, \quad i = 1, 2, \dots, p$$

Variance and covariance of the principal components are:

$$\text{Var}(Y_i) = \mathbf{e}_i' \Sigma \mathbf{e}_i = \lambda_i, \quad i = 1, 2, \dots, p$$

$$\text{Cov}(Y_i, Y_k) = \mathbf{e}_i' \Sigma \mathbf{e}_k = 0, \quad i \neq k$$

Sum of the variances of the random variables:

$$\sigma_{11} + \sigma_{22} + \dots + \sigma_{pp} = \sum_{i=1}^p \text{Var}(X_i) = \lambda_1 + \lambda_2 + \dots + \lambda_p = \sum_{i=1}^p \text{Var}(Y_i)$$

Proportion of total population variance due to the k -th principal component:

$$(\text{Variance explained by } k) = \frac{\lambda_k}{\lambda_1 + \lambda_2 + \dots + \lambda_p}$$

Correlation between the i -th principal component Y_i and the k -th variable X_k :

$$\rho_{Y_i, X_k} = \frac{e_{ik} \sqrt{\lambda_i}}{\sqrt{\sigma_{kk}}}, \quad i, k = 1, 2, \dots, p$$

```
princomp(x = head_dat, scores=TRUE)
princomp(covmat = sigma, scores=TRUE)
```


5.3 PCA with Correlation

$$y_i = \mathbf{e}_i' \mathbf{z} = e_{i1}z_1 + e_{i2}z_2 + \cdots + e_{ip}z_p, \quad i = 1, 2, \dots, p$$

$$\text{Var}(y_i) = \lambda_i, \quad i = 1, 2, \dots, p$$

$$\text{Cov}(y_i, y_k) = 0, \quad i \neq k$$

$$(\text{Proportion of standardised variance explained by } i\text{th pc}) = \frac{\lambda_i}{p}$$

$$r_{y_i, z_k} = e_{ik} \sqrt{\lambda_i}, \quad i, k = 1, 2, \dots, p$$

```
princomp(x = head_dat, cor=TRUE, scores=TRUE)
princomp(covmat=cor.matrix, scores=TRUE)
```

5.4 PCA Plots

```
screeplot(pca, type="barplot")
screeplot(pca, type="lines")
biplot(pca)
```

6 Factor Analysis

6.1 Purpose

- To reduce the dimensionality of large datasets.
- To uncover the underlying relationships between observed variables and latent factors.
- To summarize correlated variables using a smaller number of latent factors.

6.2 Orthogonal Factor Model

$$X - \mu = LF + \epsilon$$

Assumes:

$$E(\mathbf{F}) = 0, \text{Cov}(\mathbf{F}) = E[\mathbf{F}\mathbf{F}'] = I$$

$$E(\epsilon) = 0, \text{Cov}(\epsilon) = E[\epsilon\epsilon'] = \psi$$

And that $\mathbf{F}$ and ϵ are independent, so:

$$\text{Cov}(\epsilon, \mathbf{F}) = E[\epsilon\mathbf{F}'] = 0$$

6.2.1 Communalities and Specific Variance

The variance of an observed variable X_i is split into **communality** and **specific variance**:

$$\sigma_{ii} = l_{i1}^2 + l_{i2}^2 + \cdots + l_{im}^2 + \psi_i$$

Where:

- $\sigma_{ii} = \text{Var}(X_i)$ is the total variance of the variable.
- $l_{i1}, l_{i2}, \dots, l_{im}$ are the **factor loadings** for the i -th variable on each of the m factors.
- ψ_i is the **specific variance** (unique variance) for the i -th variable.

Alternatively, we can express **communality** h_i^2 as:

$$h_i^2 = l_{i1}^2 + l_{i2}^2 + \cdots + l_{im}^2$$

Thus, the total variance of the variable can also be written as:

$$\sigma_{ii} = h_i^2 + \psi_i, \quad i = 1, 2, \dots, p$$

Where h_i^2 is the communality, representing the portion of the variance explained by the common factors, and ψ_i is the specific variance.

$$(\text{Proportion of total sample variance due to } j\text{-th factor}) = \frac{e_{1j}^2 + e_{2j}^2 + \cdots + e_{pj}^2}{\sigma_{11} + \sigma_{22} + \cdots + \sigma_{pp}}$$

6.3 Types of Factor Analysis

6.3.1 Principal Component Factor Analysis (PCFA)

- Assume only Σ is known and we calculate A as in PCA and,
- Estimate the diagonal error matrix Ω
- It focuses on finding components that explain the maximum variance in the data.
- $\Sigma \rightarrow A = e_k \lambda_k^{1/2} \rightarrow \Omega = \Sigma_{diag} - (AA^T)_{diag}$

```
PCA = prcomp(data, scale. = TRUE )
A = PCA$rotation %*% diag( PCA$sdev )
Omega = diag(var(data))-diag(A%*%t(A))

# OR

sigma = var(data)
PCA = principal(sigma,2,rotate="none")
A = PCA$loadings
```

6.3.2 Principal Axis Factoring (PAF)

- Assume we know Σ and Ω
- Difference is an estimate of the covariance matrix Σ_A
- Calculate A based on Σ_A
- PAF focuses on explaining the shared variance (common variance) among the observed variables, excluding the unique variance.
- $\Sigma, \Omega \rightarrow \Sigma_A = \Sigma - \Omega \rightarrow A = e_{A,k} \lambda_{A,k}^{1/2}$

```
S = cov(data)
eig = eigen(S)
sigma_hat_sq = mean( eig$values[3:13] )
Om = diag( rep( sigma_hat_sq, 13 ) )
S_A = S - Om
eig_A = eigen( S_A )
Gamma_hat_2 = eig_A$vectors[,1:2]
Lambda_hat_2 = diag( eig_A$values[1:2]^(1/2) )
Ahat = Gamma_hat_2 %*% Lambda_hat_2
```

6.3.3 Maximum Likelihood Estimation (MLE)

F and ϵ are assumed to be normally distributed. When they are jointly normal the MLE's are $\hat{L}$, $\hat{\Psi}$, and $\mu - \bar{x}$

```
mle.FA <- factanal(factors = 1, covmat = ability.cov, method = "mle", scores = "regression")
```

6.4 Factor Rotation

After calculating the factor loadings, **rotation** can be applied to make the structure more interpretable. There are two main types of rotation:

- **Orthogonal Rotation** (e.g., **Varimax**): Keeps the factors uncorrelated.
- **Oblique Rotation** (e.g., **Promax**): Allows the factors to be correlated.

```
factanal( wine0, 2, rotation = "promax", scores = "regression" )  
factanal( wine0, 2, rotation = "varimax", scores = "regression" )  
principal(Harman74.cor$cov,4,rotate="varimax")
```

6.5 Biplots

```
biplot( mle.plain$scores[,1:2], mle.plain$loadings[,1:2], col = c("white", "blue") )
biplot( mle.vari$scores[,1:2], mle.vari$loadings[,1:2], col = c("white", "blue") )
biplot( PCFA$x, A.plain[,1:2], col = c("white", "blue") )
biplot( PCFA$x, A.vari$loadings[,1:2], col = c("white", "blue") )
```

7 Canonical Correlation Analysis

CCA is used to measure the relationship between two sets of variables, X and Y . The goal is to find linear combinations of the variables in each set that have the maximum correlation.

7.1 Canonical Correlation

The canonical correlation ρ between two linear combinations U and V is defined as:

$$\rho = \frac{\text{Cov}(U, V)}{\sqrt{\text{Var}(U)}\sqrt{\text{Var}(V)}}$$

Where:

- $U = a'X$ (linear combination of the variables in X).
- $V = b'Y$ (linear combination of the variables in Y).

7.2 Canonical Variates

The canonical variates are the linear combinations of the variables in X and Y :

$$U = a'X \quad \text{and} \quad V = b'Y$$

Where a and b are the vectors of weights (coefficients) for the variables in X and Y , respectively.

7.3 Sample Canonical Correlations

Let $\hat{\rho}_1^2 \geq \hat{\rho}_2^2 \geq \dots \geq \hat{\rho}_p^2$ be the ordered eigenvalues of the matrix $S_{11}^{-1/2} S_{12} S_{22}^{-1} S_{21} S_{11}^{-1/2}$, with corresponding eigenvectors $\hat{e}_1, \hat{e}_2, \dots, \hat{e}_p$. Let $\hat{f}_1, \hat{f}_2, \dots, \hat{f}_p$ be the eigenvectors of $S_{22}^{-1/2} S_{21} S_{11}^{-1} S_{12} S_{22}^{-1/2}$.

Then the k -th sample canonical variate pair is:

$$\begin{aligned} \hat{U}_k &= \hat{e}_k S_{11}^{-1/2} x^{(1)} \rightarrow \hat{a}_k = \hat{e}_k S_{11}^{-1/2} \\ \hat{V}_k &= \hat{f}_k S_{22}^{-1/2} x^{(2)} \rightarrow \hat{b}_k = \hat{f}_k S_{22}^{-1/2} \end{aligned}$$

Where $x^{(1)}$ and $x^{(2)}$ are the values of the variables $X^{(1)}$ and $X^{(2)}$ for a particular experimental unit. Also, the first sample canonical variate pair has the maximum sample correlation.

For the k -th pair, the sample correlation is:

$$r_{\hat{U}_k, \hat{V}_k} = \hat{\rho}_k^*$$

The quantities $\hat{\rho}_1^*, \hat{\rho}_2^*, \dots, \hat{\rho}_p^*$ are the **sample canonical correlations**.

```
library(CCA)
cca.1 <- cc(X1, X2)

CCA = cancelor(XX1.s, XX2.s)
```

- cor: the rho_hats, the CC sample correlations
- xcoef
- ycoef
- scores: canonical variates
- scores\$yscores: Y canonical variates
- scores\$xscores: X canonical variates
- scores\$corr.X.xscores: Correlation between X and X canonical variates (the U's)
- scores\$corr.Y.xscores: Correlation between Y and X canonical variates
- scores\$corr.X.yscore: Correlation between X and Y canonical variates (the V's)
- scores\$corr.Y.yscore: Correlation between Y and Y canonical variates

7.4 Hypothesis Testing in CCA

To test the hypothesis that the canonical correlations are zero:

1. Null hypothesis: $H_0 : \rho = 0$
2. Test statistic:

$$-2\log(\Lambda) = -2\log\left(\prod_{i=1}^k (1 - \rho_i^2)\right)$$

Where ρ_i are the canonical correlations. 3. The test statistic follows a chi-squared distribution.

```
Tk = function(k, n, d1, d2, vv) {
  rr = length(vv)
  Tkout = -(n - (d1 + d2 + 3)/2) * log(prod(1 - vv[(k + 1):rr]^2))
  dof = (d1 - k) * (d2 - k)
  pval = pchisq(Tkout, df = dof, lower.tail = FALSE)
  list(Tkout = Tkout, pval = pval)
}
```

```
Tk(k = 1, n = 709, d1 = 3, d2 = 3, vv = CCA$cor)
```

```
library(yacca)
cca.fit = cca(air.rearranged[, 1:3], air.rearranged[, 4:6])
summary(cca.fit)
```

8 Cluster Analysis

- Identify natural groupings: uncover inherent groupings within data
- Data-driven approach: based on proximity, typically using distance or similarity measures, with minimal prior assumptions.
- Reveal underlying structure: expose hidden patterns

8.1 Measures of Association

Proximity Measures:

- Similarity measure $s(x_i, x_k)$ (large if x_i, x_k is similar)
- Distance measure $d(x_i, x_k)$ (small if x_i, x_k is similar)

Two Common Measures We Need:

- Measure of association between Sample & Units
- Measure of association between Clusters

8.1.1 Continuous Variables

The measure of association must satisfy the following properties:

Let $x, y, z \in \Omega$:

- i) *Identity* $d(x, y) = 0$ if $x=y$ element wise
- ii) *Symmetry* $d(x, y) = d(y, x)$
- iii) *Triangle Inequality* $d(x, z) + d(z, y) \geq d(x, y)$
- iv) *Positivity* $d(x, y) \geq 0$

Statistical Distance

$$d(x, y) = \sqrt{(x - y)' S^{-1} (x - y)}$$

Euclidean Distance

$$d(x, y) = \sqrt{(x - y)' (x - y)}$$

8.1.2 Binary Variables

- These are variables that take on only two possible values, typically 0 or 1, indicating the absence or presence of a characteristic
- The goal is to determine how “similar” two items (or observations) are based on these binary variables

Similarity vs. Dissimilarity

- If two items have more binary variables in common (e.g., both have 1's in the same positions), they are considered more similar.
- If their binary variables differ a lot (e.g., one has a 1 where the other has a 0), they are considered less similar or more dissimilar.

- Often the 1 to 1 match is a stronger indication of similarities than 0 to 0 match.

	$k = 1$	$k = 0$	Totals
$i = 1$	a	b	$a + b$
$i = 0$	c	d	$c + d$
Totals	$a + c$	$b + d$	$n = a + b + c + d$

Weighted Similarity

- Sometimes, not all matches are considered equal.
- See lecture slides for weights

Coefficient	Rationale
$\frac{a+d}{p}$	Equal weights for 1-1 matches and 0-0 matches.
$\frac{2(a+d)}{2(a+d)+b+c}$	Double weight for 1-1 matches and 0-0 matches.
$\frac{a+d}{a+d+2(b+c)}$	Double weight for unmatched pairs.
$\frac{a}{p}$	No 0-0 matches in numerator.

Sample Correlation Coefficient

$$r = \frac{ad - bc}{\sqrt{(a+b)(c+d)(a+c)(b+d)}}$$

8.2 Hierarchical

Hierarchical methods can be adapted to cluster variables rather than observations.

8.2.1 Agglomerative hierarchical algorithms

- Start by defining each data point as a cluster;
- Then, the two closest clusters are combined into a new cluster;
- In each subsequent step, two existing clusters are merged into a single cluster.

Single Linkage Focus on the closest element in the cluster to the elements outside of the cluster

$$d_{A,B} = \min(d_{ij})$$

$$d[(xy), z] = \min\{d(x, z), d(y, z)\}$$

$$\begin{aligned} d[(xy), (uv)] &= \min\{d[x, (uv)], d[y, (uv)]\} \\ &= \min\{d(x, u), d(x, v), d(y, u), d(y, v)\} \end{aligned}$$

```
df <- data.frame(data)

# On observations
dist <- dist(df)
complete = hclust(dist, method ="single")

# On variables
dist <- as.dist(1-cor(df))
complete = hclust(dist, method ="single")
```


Complete Linkage Focus on the element in the cluster that is furthest away

$$d_{A,B} = \max(d_{ij})$$

```
df <- data.frame(data)

# On observations
dist <- dist(df)
complete = hclust(dist, method = "complete")

# On variables
cor.dist <- as.dist(1-cor(df))
complete = hclust(cor.dist, method = "complete")
```

8.2.2 Divisive hierarchical algorithms

- Start by putting all data points into a single cluster;
- Then we divide this cluster into two clusters;
- At each subsequent step, we divide an existing cluster into two clusters.

$$d[(UV), W] = \max\{d(U, W), d(V, W)\}$$

8.2.3 Useful Functions

```
cutree(hc, 2)
plot(hc, main="Dendrogram", xlab="", sub="", cex = .6)
```

8.3 Non-hierarchical

Hierarchical methods start by assigning data points into a predefined number of clusters, then iteratively adjust the clusters

8.3.1 K-means Clusters

Objective: The main goal is to partition a dataset into K clusters such that the sum of squared distances from each data point to the nearest cluster centroid is minimised.

Method:

1. Start with K initial clusters (either randomly or based on some “good” guess).
2. Assign each point to the nearest centroid (using Euclidean distance).
3. Recalculate the centroids (mean) of the clusters.
4. Repeat steps 2 and 3 until the assignments stabilise and no further points change clusters.

```
# Not scaled
df <- data.frame(data)
scaled.kmean <- kmeans(df, centers = 5, nstart = 25)

# Scaled
df.scaled <- scale(df)
scaled.kmean <- kmeans(df.scaled, centers = 5, nstart = 25)
```

Elbow Method: Finding Optimal Number of Clusters Plotting the Total Within Sum of Squares against the number of clusters and finding the “elbow” point where the rate of decrease slows. This point indicates a good trade-off between reducing variance and the number of clusters.

1. Loop through $k=1,2,3,\dots,n$
2. Get Total Within SS
3. Plot trade-off
4. Pick best k (at “elbow”)

Can R to find best k : - *Silhouette method* - *Gap statistic*

```
# Elbow Graphs
fviz_nbclust(df_scaled, kmeans, method = "wss")
fviz_nbclust(df_scaled, kmeans, method = "silhouette")
fviz_nbclust(df_scaled, kmeans, nstart = 20, method = "gap_stat", nboot = 50)

# Cluster Graph
fviz_cluster(k2, geom = "point", data = df)
```

8.4 Agglomerative vs k-means

	AH	k-means
hierarchical	yes	no
outliers	easy to detect	allocated to clusters
# of clusters	not determined	user provides
cluster allocation	same each run	depends on initial cluster centres
reproducible	yes	no
optimal arrangements	no	yes
visual tools	dendrograms	within-cluster variability plots

9 Linear Discriminant Analysis

- Discriminant analysis is a technique used to characterize the differences among the groups of objects or observations.
- This characterization can also be used to allocate new objects to pre-defined group.

9.1 Discrimination and Classification

Discrimination

Concerned with separating distinct sets of objects.

- Rather exploratory in nature
- Often employed on a one-time basis in order to investigate observed differences when causal relationships are not well understood.

Classification

Concerned with allocating new objects to previously defined groups.

- Less exploratory in the sense that they lead to well-defined rules.
- Ordinarily requires more problem structure than discrimination does.

9.2 Discrimination and Two Population Structure

PDFs

- $\pi_1 : f_1(x)$
- $\pi_2 : f_2(x)$

Sample Space Ω

- R_1 , set of values x for which we classify object as π_1
- $R_2 = \Omega - R_1$, set of values x for which we classify object as π_2

Optimal Classification

- Expected Cost Misclassification (ECM)
- Total Probability Misclassification (TPM)

Probability of Misclassification

Prior Probability:

- $\pi_1 : p_1$
- $\pi_2 : p_2$

$$P(\text{Observation misclassified as } \pi_1) = p_2 P(1|2)$$

$$P(\text{Observation misclassified as } \pi_2) = p_1 P(2|1)$$

Cost of Misclassification

- $C(1|2)$: Observation from π_2 classified as π_1
- $C(2|1)$: Observation from π_1 classified as π_2

$$\text{Expected Cost of Misclassification (ECM)} = C(2|1)P(2|1)p_1 + C(1|2)P(1|2)p_2$$

9.2.1 Regions that Minimise ECM

The regions R_1 and R_2 that minimize ECM are defined by the values of x for which:

$$R_1 : \frac{f_1(x)}{f_2(x)} \geq \frac{C(1|2) p_2}{C(2|1) p_1}$$

$$R_2 : \frac{f_1(x)}{f_2(x)} < \frac{C(1|2) p_2}{C(2|1) p_1}$$

When $C(1|2) = C(2|1)$ and $p_1 = p_2$, assign x to π_1 for:

$$f_1(x) \geq f_2(x)$$

9.3 Classification of Two Normal Populations

9.3.1 Equal Covariance Matrix

$$S_{pooled} = \frac{(n_1 - 1)S_1 + (n_2 - 1)S_2}{n_1 + n_2 - 2}$$

Allocate x_0 to π_1 if:

$$(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_{pooled}^{-1} \mathbf{x}_0 - \frac{1}{2}(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_{pooled}^{-1} (\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2) \geq \ln \left[\left(\frac{c(1|2)}{c(2|1)} \right) \left(\frac{p_2}{p_1} \right) \right]$$

9.3.1 Unequal Covariance Matrix

Allocate x_0 to π_1 if:

$$-\frac{1}{2} \mathbf{x}_0' (\Sigma_1^{-1} - \Sigma_2^{-1}) \mathbf{x}_0 + (\mu_1' \Sigma_1^{-1} - \mu_2' \Sigma_2^{-1}) \mathbf{x}_0 - k \geq \ln \left[\left(\frac{c(1|2)}{c(2|1)} \right) \left(\frac{p_2}{p_1} \right) \right]$$

```
library(car)
# No CV
Fish.lda <- lda(Origin ~ Freshwater + Marine, data = Fishesdata)
# With CV
Fish.lda.cv <- lda(Origin ~ Freshwater + Marine, data = Fishesdata, CV=TRUE)
```

9.4 Evaluating Classification Function

- Based on the following confusion matrix

Actual membership	Predicted membership		
	π_1	π_2	
π_1	n_{1C}	$n_{1M} = n_1 - n_{1C}$	n_1
π_2	$n_{2M} = n_2 - n_{2C}$	n_{2C}	n_2

9.4.1 Apparent Error Rate (APER)

- Is approximate

$$APER = \frac{n_{1M} + n_{2M}}{n_1 + n_2}$$

9.4.2 Accuracy (ACC)

$$ACC = \frac{n_{1C} + n_{2C}}{n_1 + n_2}$$

```
# No CV
T1 = table(predict(Fish.lda, type="class")$class, Fishesdata$Origin)
# With CV
T2 = table(lda.cv$class, diabetes.data$diabetes)

Accuracy = (T1[1,1]+T1[2,2])/sum(T1)
APER = (T1[1,2]+T1[2,1])/sum(T1)
```

9.5 Classification with Several Populations

$$P(\text{classifying item as } \pi_k | \pi_i) = \int_{R_k} f_i(x)$$

Conditional-classifying X from 1 into some other group:

$$ECM(1) = \sum_{k=2}^g P(k|1)C(k|1)$$

$$ECM = p_1 ECM(1) + p_2 ECM(2) + \dots + p_g ECM(g)$$

9.5.1 Rule for Classification

Classification regions that minimize the ECM are defined by allocation x to that population π_k , $k = 1, 2, \dots, g$ for which the following equation is the smallest:

$$(\bar{x}_1 - \bar{x}_2)' S_{\text{pooled}}^{-1} x_0 - \frac{1}{2} (\bar{x}_1 - \bar{x}_2)' S_{\text{pooled}}^{-1} (\bar{x}_1 + \bar{x}_2) \geq \ln \left[\frac{c(1|2) p_2}{c(2|1) p_1} \right]$$

$$(\mu_1 - \mu_2)' \Sigma^{-1} x_0 - \frac{1}{2} (\mu_1 - \mu_2)' \Sigma^{-1} (\mu_1 + \mu_2) \geq \ln \left[\frac{c(1|2)}{c(2|1)} \cdot \frac{p_2}{p_1} \right]$$

9.6 Fisher's Rule

$$\hat{y} = \hat{a}^T x = [\bar{x}_1 - \bar{x}_2] S_{\text{pooled}}^{-1} x$$

$$\bar{y}_1 = \hat{a}^T \bar{x}_1, \quad \bar{y}_2 = \hat{a}^T \bar{x}_2$$

```

Sp=((n1-1)/(n1-1+n2-1))*S1+((n2-1)/(n1-1+n2-1))*S2
invSp=solve(Sp)
xbar1=matrix(data=c(-0.135,-0.07786),nrow=2, byrow=TRUE)
xbar2=matrix(data=c(-0.308, -0.00599),nrow=2, byrow=TRUE)
xdiff=xbar1-xbar2
a=t(xdiff)%*%invSp

# Midpoint for classification
ybar1=a%*%xbar1
ybar2=a%*%xbar2
m=0.5*(ybar1+ybar2)

# Classifying -> x0 to pi1 if yhat0 > m
yhat0=a%*%x0

```

10 Logistic Regression

- For logistic regression: we're using 'natural log' or taking logs to base e .
- We are interested in modelling the relationship between a numerical explanatory variable (x) and a categorical response variable (y) with only two outcomes.
- We model the probability (p) of how the outcomes relate to the explanatory variable (x).
- "logit function" or "log odds", which is the $\log_e$ of the odds

10.1 Logit Function

- Logistic regression works with odds, $Odds = \frac{p}{1-p}$.

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x$$

10.2 Modelling

```
glm(Signs ~ Age, family = "binomial", data = gforces)
```

10.2.1 Interpretation

An increase of 1 unit in x_i will increase (if +ve) or decrease (if -ve) the odds of y by e^{b_i}

$$p = \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}}$$

10.2.2 Hypothesis Testing

$$H_0 : \beta_1 = 0 \text{ vs } H_1 : \beta_1 \neq 0$$

Test Statistic

$$z = \frac{\beta_1}{SE(\beta_1)}$$

p-value

$$z \sim N(0, 1)$$

```
z = abs(b1 / b1.se)
critical = qnorm(0.05, 0, 1, lower.tail = FALSE)
# OR
p-val = pnorm(z, 0, 1, lower.tail = FALSE)
```

10.2.3 Confidence Interval

CI for β_1

$$\beta_1 \pm z^* SE(\beta_1)$$

CI for the Odds Ratio

$$e^{\beta_1 \pm z^* SE(\beta_1)}$$

10.2.4 Goodness of Fit - Residual Deviance

- Aims to check whether the model with predictors fits significantly better than a model with just an intercept (i.e. null model).
- The test statistic is distributed chi-squared with degrees of freedom equal to the differences in degrees of freedom between the current and the null model (i.e. number of predictor variables in the model)

```
pchisq(q = 2.1348, df = 1, lower.tail=FALSE)
```

10.2.4 Forward Selection

1. Fit model with intercept term only
2. Iteratively add the variable with the smallest model AIC
3. Repeat until it becomes worse

```
Heart.0 <- glm(AHD ~ 1, data = Train, family = binomial)
Heart.all <- glm(AHD ~ ., data = Train, family = binomial)
Heart.fwd <- step(Heart.0, scope = formula(Heart.all), direction = "forward", trace = 0)
```

10.3 Confusion Matrix

		True condition			
Total population		Condition positive	Condition negative	Prevalence $= \frac{\Sigma \text{Condition positive}}{\Sigma \text{Total population}}$	Accuracy (ACC) = $\frac{\Sigma \text{True positive} + \Sigma \text{True negative}}{\Sigma \text{Total population}}$
Predicted condition	Predicted condition positive	True positive	False positive, Type I error	Positive predictive value (PPV), Precision = $\frac{\Sigma \text{True positive}}{\Sigma \text{Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{False positive}}{\Sigma \text{Predicted condition positive}}$
	Predicted condition negative	False negative, Type II error	True negative	False omission rate (FOR) = $\frac{\Sigma \text{False negative}}{\Sigma \text{Predicted condition negative}}$	Negative predictive value (NPV) = $\frac{\Sigma \text{True negative}}{\Sigma \text{Predicted condition negative}}$
		True positive rate (TPR), Recall, Sensitivity, probability of detection, $\text{Power} = \frac{\Sigma \text{True positive}}{\Sigma \text{Condition positive}}$	False positive rate (FPR), Fall-out, probability of false alarm $= \frac{\Sigma \text{False positive}}{\Sigma \text{Condition negative}}$	Positive likelihood ratio (LR+) $= \frac{\text{TPR}}{\text{FPR}}$	Diagnostic odds ratio (DOR) $= \frac{\text{LR+}}{\text{LR-}}$ $2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$
		False negative rate (FNR), Miss rate $= \frac{\Sigma \text{False negative}}{\Sigma \text{Condition positive}}$	Specificity (SPC), Selectivity, True negative rate (TNR) $= \frac{\Sigma \text{True negative}}{\Sigma \text{Condition negative}}$	Negative likelihood ratio (LR-) $= \frac{\text{FNR}}{\text{TNR}}$	

Figure 1: Confusion Matrix

11 Principal Component Regression

Aim: Potential problem of multicollinearity as evidenced of the very high correlation

```
pca <- princomp(data, cor = TRUE)
reg <- lm(Y ~ pca$scores, data = data)
```

12 Extra

12.1 Code

```
library(PerformanceAnalytics)
chart.Correlation(sweat, histogram=TRUE, pch=19)

ggpairs(iris, aes(colour=Species, alpha=0.4))
```

```
sweat = read.csv("sweatdata.csv")
means = colMeans(sweat)
sigma = var(sweat)
means
round(sigma,3)
```

```
data(Boston)
boston = Boston[,-4]
scaled.bost = scale(boston)
mle.plain <- factanal(x=scaled.bost, factors = 3, method = "mle", scores="regression",
                     rotation = "none")
mle.vari <- factanal(x=scaled.bost, factors = 3, method = "mle", scores="regression")

h.plain=0
h.vari=0
for (i in 1:13)
{
  h.plain[i]=sum(mle.plain$loadings[i,]^2)
  h.vari[i]=sum(mle.vari$loadings[i,]^2)
}
mle.plain.loadings = mle.plain$loadings
mle.plain.uniqueness = mle.plain$uniquenesses
mle.plain.communalities = h.plain

mle.vari.loadings = mle.vari$loadings
mle.vari.uniqueness = mle.vari$uniquenesses
mle.vari.communalities = h.vari
```

```
sigma.bost = var(boston)
PCFA = prcomp(boston, scale. = TRUE )

A.plain = (PCFA$rotation %*% diag( PCFA$sdev ))[,1:3]
A.vari = varimax((PCFA$rotation %*% diag( PCFA$sdev ))[,1:3])
Omega.plain = diag(sigma.bost)-diag(A.plain%*%t(A.plain))
Omega.vari = diag(sigma.bost)-diag(A.vari$loadings%*%t(A.vari$loadings))

h.PCFA.plain=0
h.PCFA.vari=0
for (i in 1:13)
{
  h.PCFA.plain[i]=sum(A.plain[i,]^2)
  h.PCFA.vari[i]=sum(A.vari$loadings[i,]^2)
}
```

```

plain.loadings = A.plain
plain.uniqueness = Omega.plain
plain.communalities = h.PCFA.plain

vari.loadings = A.vari$loadings
vari.uniqueness = Omega.vari
vari.communalities = h.PCFA.vari

```

12.2 Written

1. Given a correlation matrix, one can obtain corresponding covariance matrix.

False. If given a correlation matrix, you cannot obtain the corresponding covariance matrix. The correlation matrix is on a standardised scale and so the variance of the variables is not known. For example, the formula for correlation is $Corr(X, Y) = \frac{Cov(X, Y)}{SD(X)SD(Y)}$ which when rearranged gives $Cov(X, Y) = Corr(X, Y)SD(X)SD(Y)$. So to re construct the covariance matrix you need to know the variances of each of the variables.

2. The orthogonal factor model with m common factors has one of the covariances structures, that is, $Cov(L, F) = X$, where X is the observable random vector with p components, L be the matrix of factor loadings and F be the vector of m -factors $F_1, F_2, \dots, F_m$.

False, $Cov(L, F) = X$, does not hold. The factors, F , are assumed to be normally distributed with mean 0 and variance I , $F \sim N(0, I)$. The loadings are fixed and deterministic and are not randomly distributed. Therefore, the $Cov(L, F)$ would be undefined. Moreover, the dimensions of the two matrices are different so no attempt at computing a covariance would give a result.

3. What is the main focus of supervised dependence analysis in multivariate regression analysis?

Investigating relationships between relevant variables

4. To visualise multivariate continuous data, one can use.

A scatterplot matrix